

Week 1 Data Quality Assessment Report

1. Overview

This report summarizes the initial data quality assessment for the raw datasets collected in Week 1. The assessment focuses on dataset size, field coverage, missing values, duplicate records, and key field availability.

2. Database Summary

Table	Rows	Columns	Duplicate Rows
iea_ev_data_explorer_2026_raw	49176	9	15
kaggle_global_ev_charging_station_raw	257585	19	3
openchargemap_charging_stations_sample	133026	17	223
tesla_deliveries_2015_2025_raw	2640	12	0
us_alt_fuel_stations_historical_raw	95300	75	0
world_bank_gdp_country_raw	266	71	0
world_bank_income_level_country_raw	265	6	0
world_bank_population_raw	17191	41	0
world_bank_road_density_raw	1103	38	0

3. Visual Summary

The following figures provide a quick visual summary of dataset size, duplication, field coverage, and missing-value patterns.

Figure 1. Row Counts by Raw Data Table

This figure compares the number of records across all raw tables. It helps identify which data sources contribute the largest amount of data.

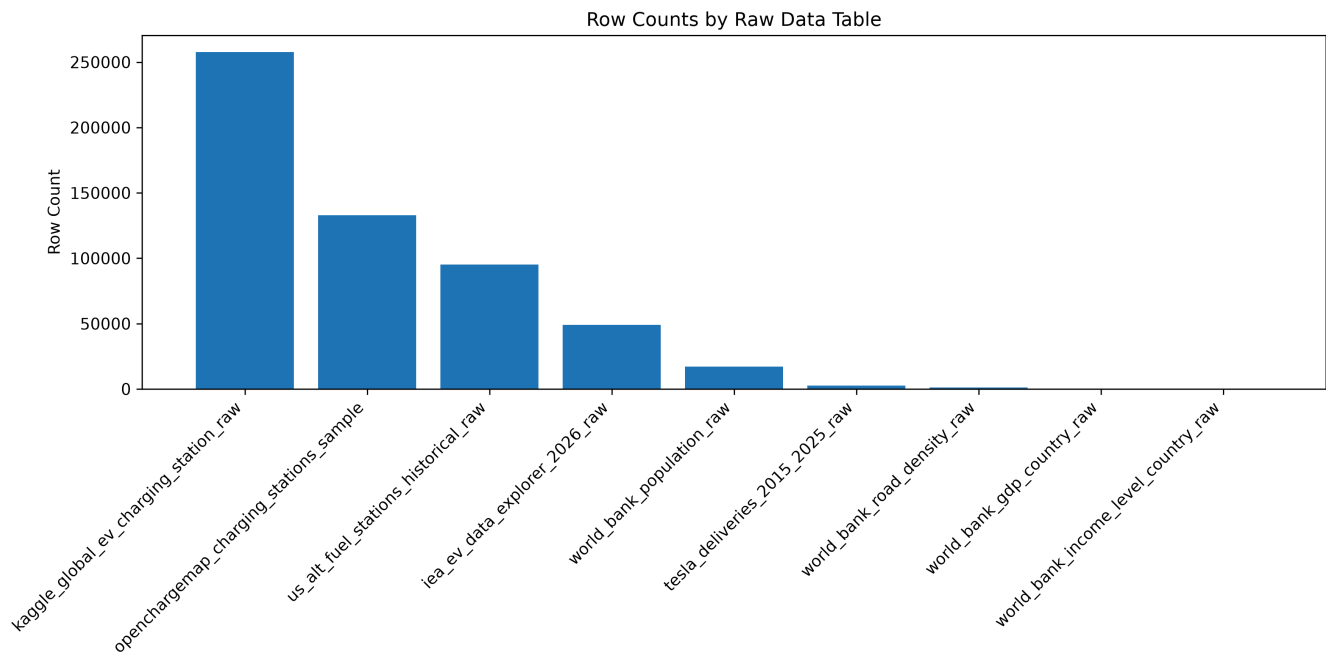


Figure 2. Duplicate Rows by Raw Data Table

This figure shows the number of fully duplicated rows in each raw table. Tables with duplicate rows will need further review during Week 2 cleaning.

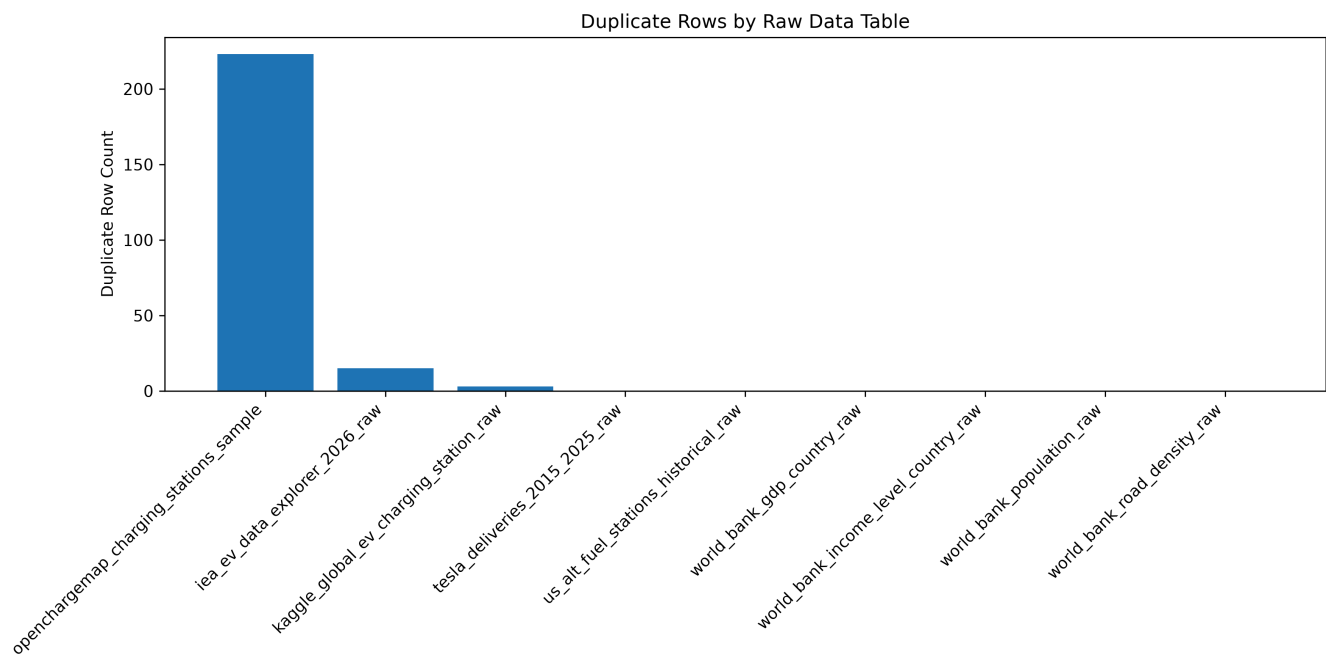
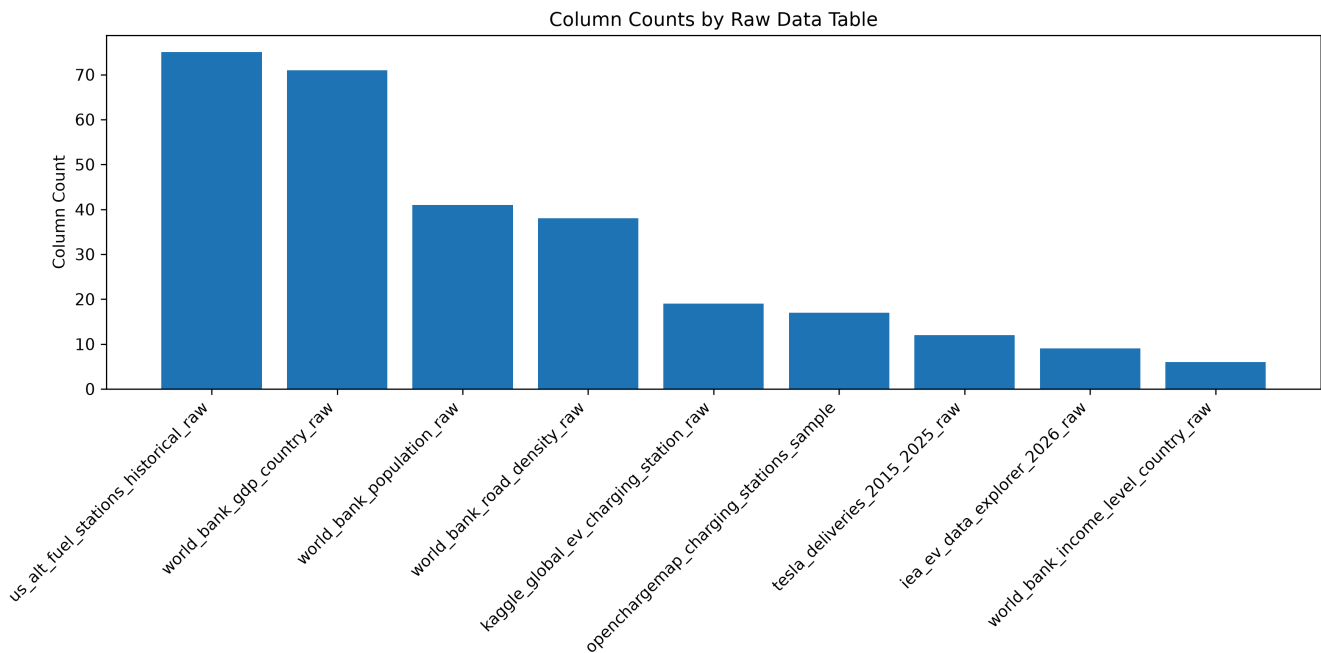


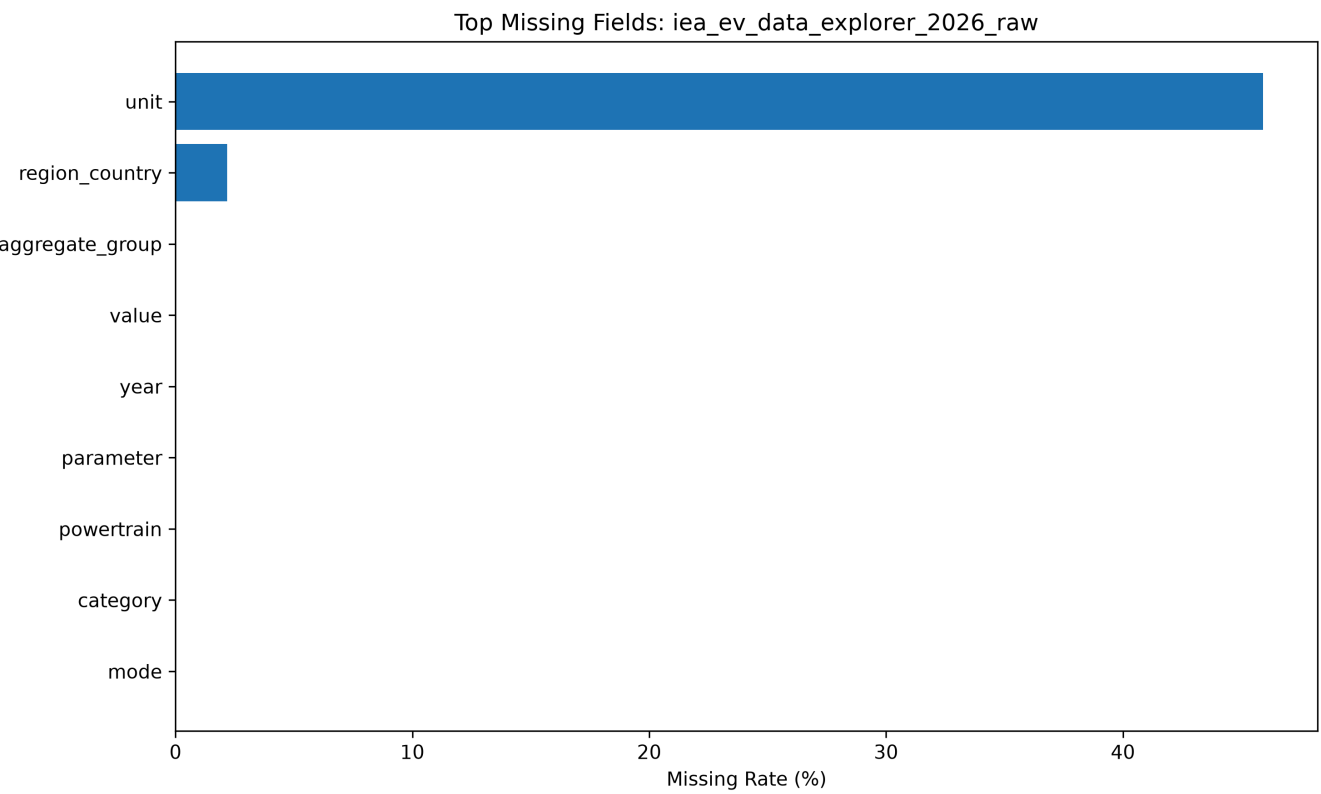
Figure 3. Column Counts by Raw Data Table

This figure compares the number of fields available in each raw table. Tables with more columns may require more careful field selection and standardization.



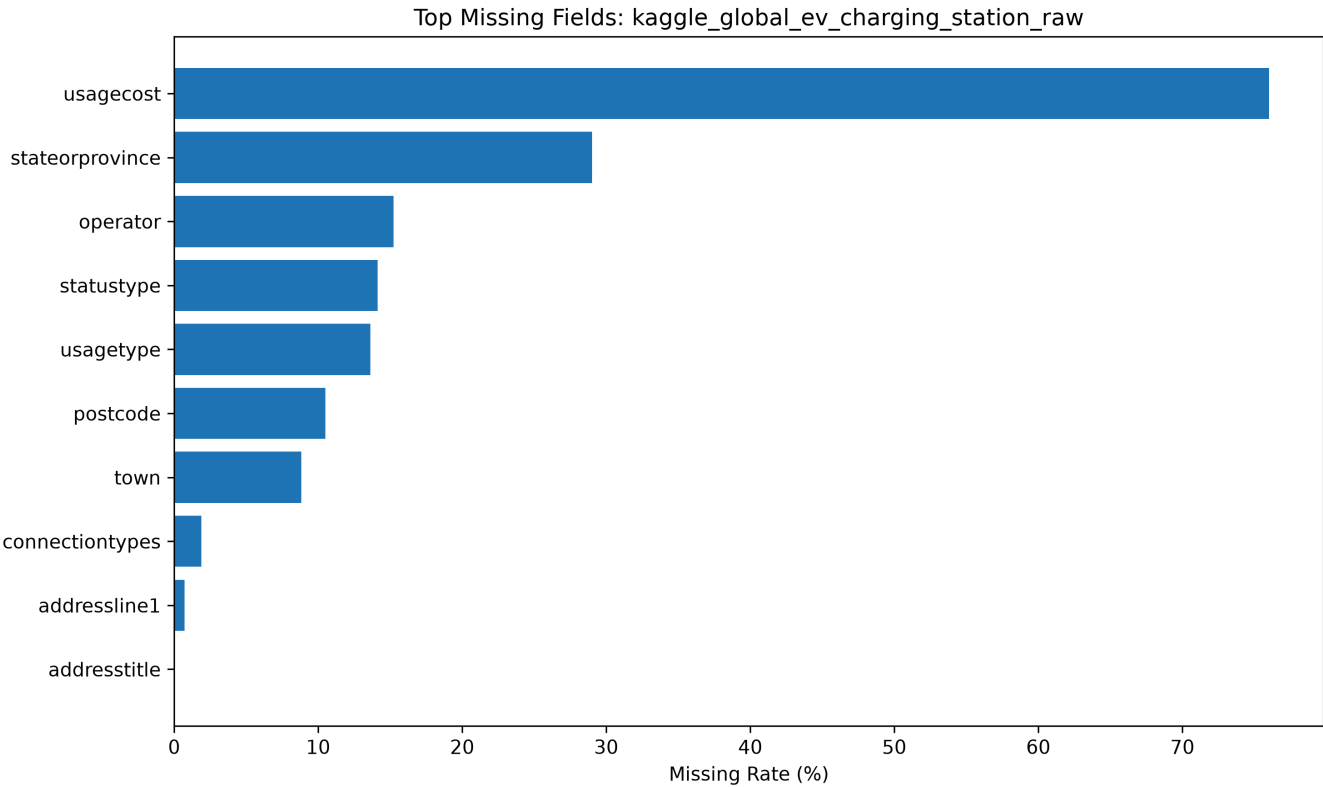
Missing Values: `iea_ev_data_explorer_2026_raw`

This figure shows the top fields with the highest missing-value rates in `iea_ev_data_explorer_2026_raw`.



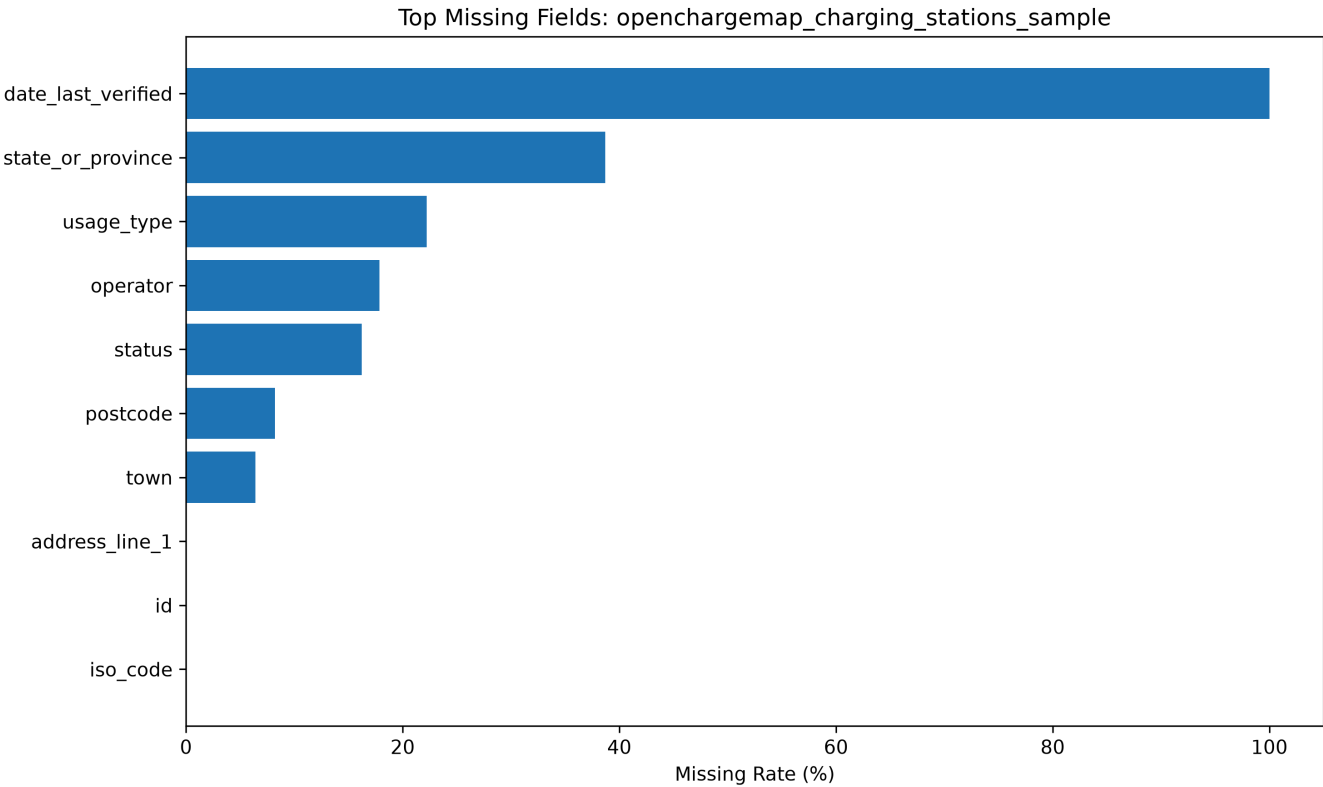
Missing Values: `kaggle_global_ev_charging_station_raw`

This figure shows the top fields with the highest missing-value rates in `kaggle_global_ev_charging_station_raw`.



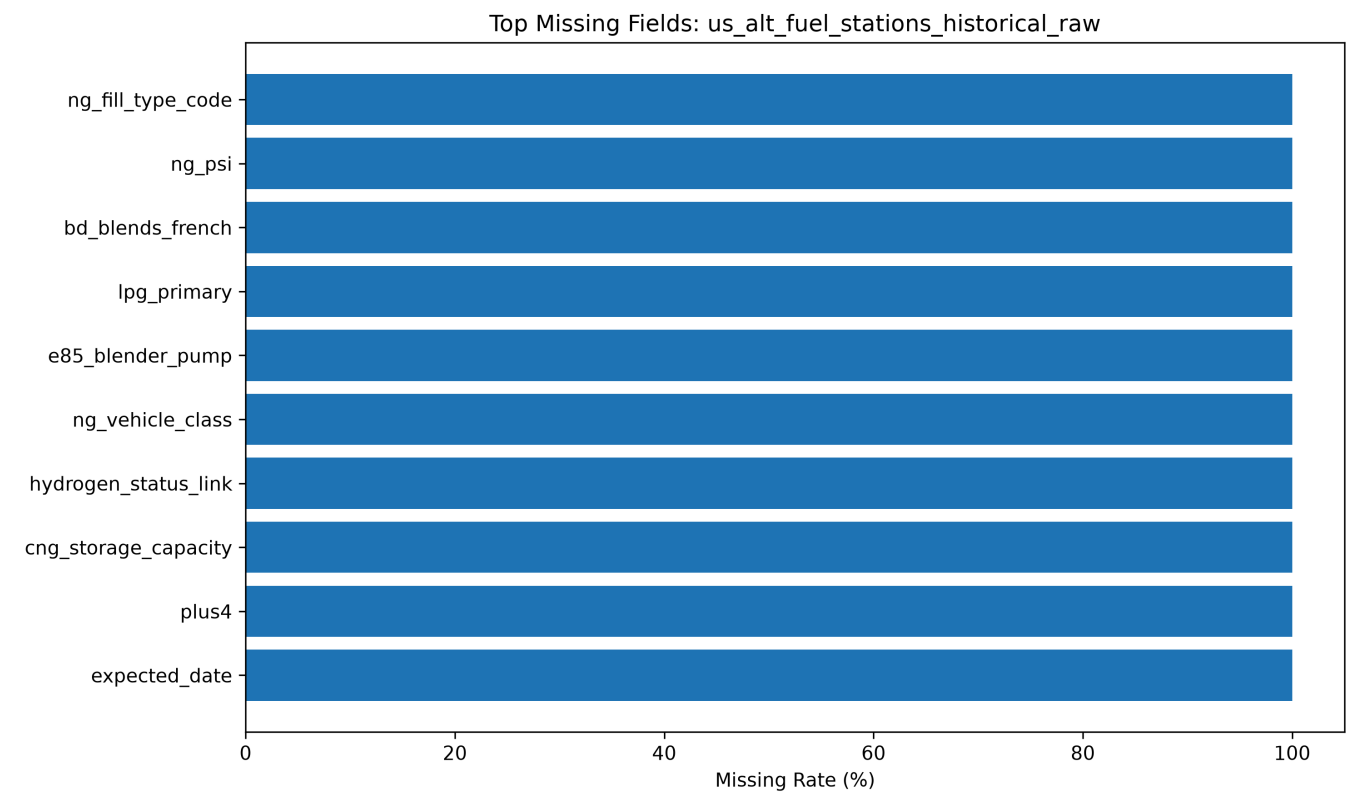
Missing Values: openchargemap_charging_stations_sample

This figure shows the top fields with the highest missing-value rates in `openchargemap_charging_stations_sample`.



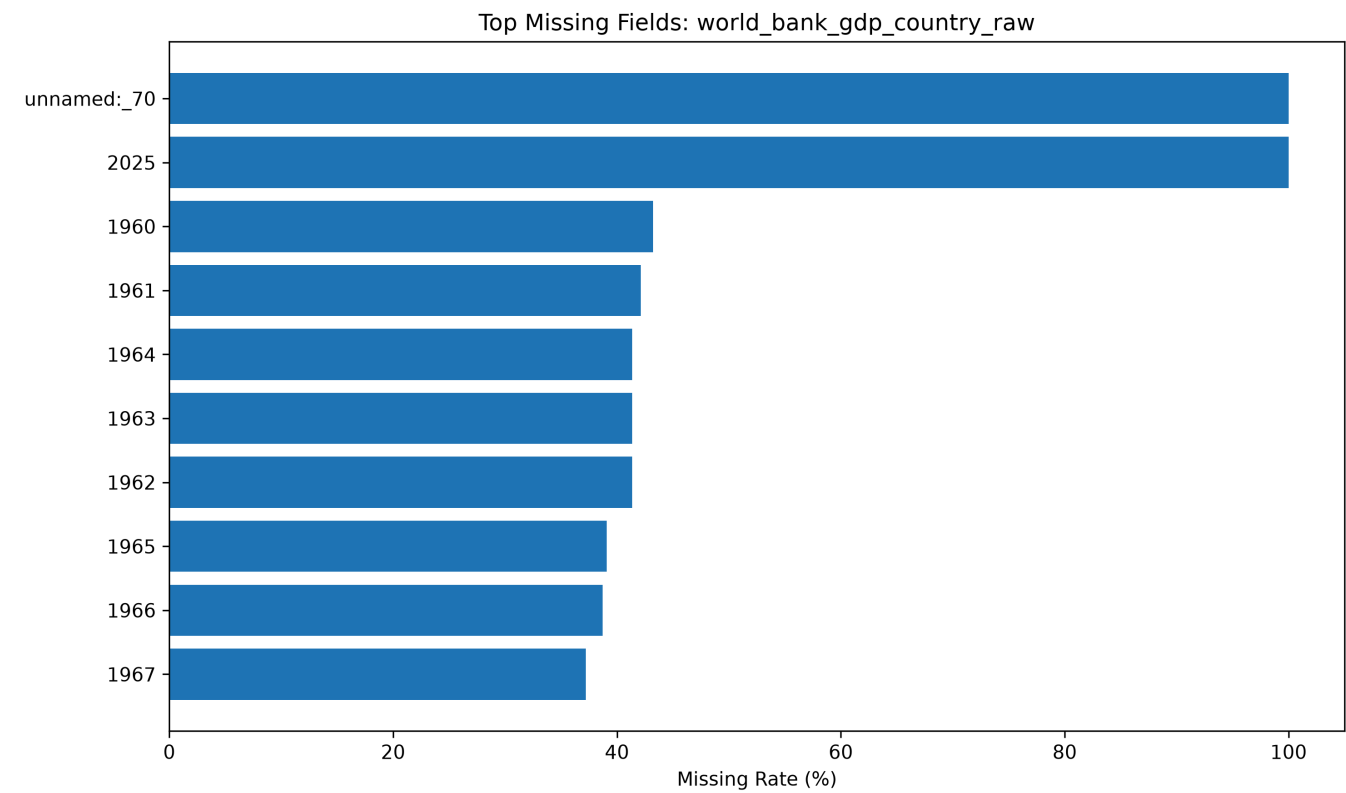
Missing Values: us_alt_fuel_stations_historical_raw

This figure shows the top fields with the highest missing-value rates in `us_alt_fuel_stations_historical_raw`.



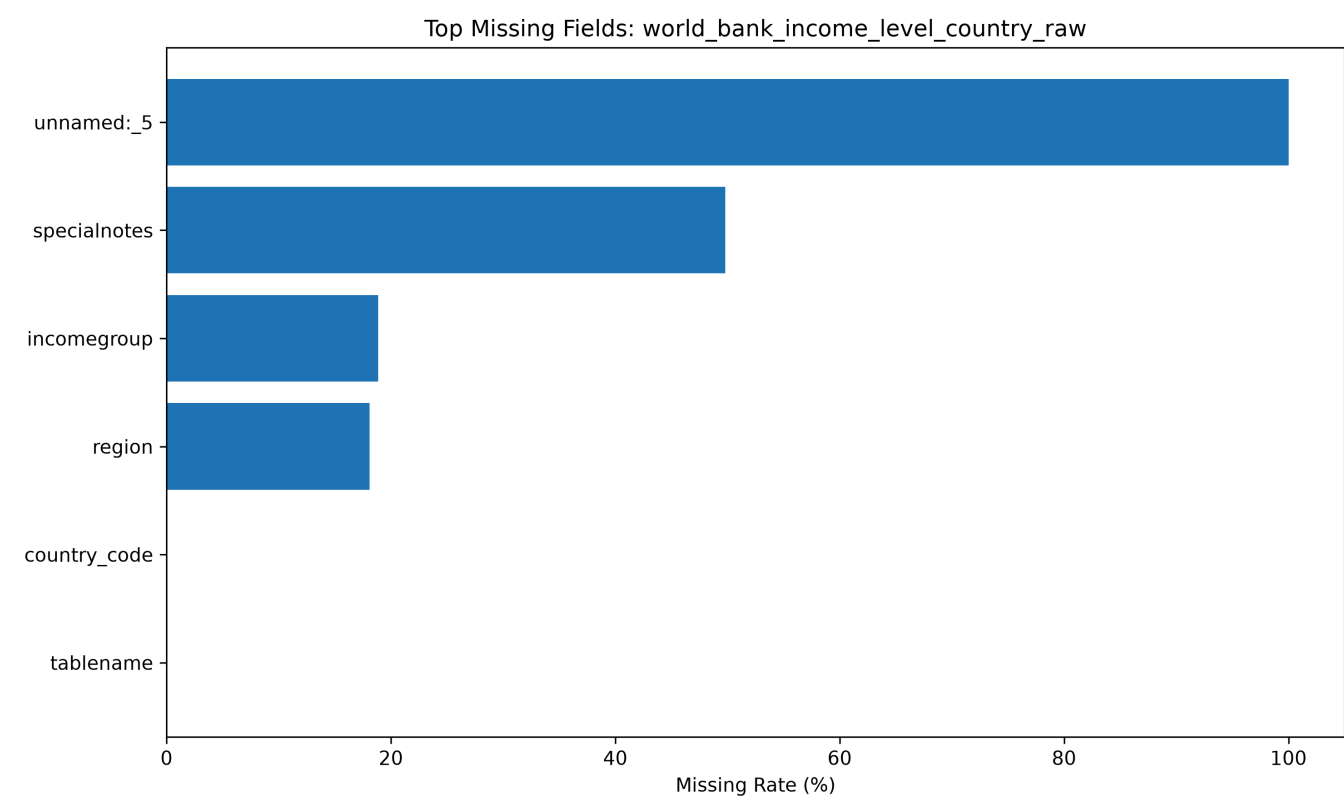
Missing Values: `world_bank_gdp_country_raw`

This figure shows the top fields with the highest missing-value rates in `world_bank_gdp_country_raw`.



Missing Values: `world_bank_income_level_country_raw`

This figure shows the top fields with the highest missing-value rates in `world_bank_income_level_country_raw`.



4. Table-Level Quality Checks

4.1 ie_a_ev_data_explorer_2026_raw

- Rows: 49176
- Columns: 9
- Duplicate rows: 15

Column names:

region_country, category, parameter, mode, powertrain, year, unit, value, aggregate_group

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
unit	22582	45.92
region_country	1078	2.19
category	0	0.0
mode	0	0.0
parameter	0	0.0

Field	Missing Count	Missing Rate (%)
powertrain	0	0.0
year	0	0.0
value	0	0.0
aggregate_group	0	0.0

4.2 kaggle_global_ev_charging_station_raw

- Rows: 257585
- Columns: 19
- Duplicate rows: 3

Column names:

stationid, uuid, dataproviderid, operator, usagetype, usageecost, addresstitle, addressline1, town, stateorprovince, postcode, country, latitude, longitude, maxpowerkw, fastchargecount, connectiontypes, statustype, yearcreated

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
usageecost	195766	76.0
stateorprovince	74752	29.02
operator	39225	15.23
statustype	36392	14.13
usagetype	35116	13.63
postcode	27075	10.51
town	22756	8.83
connectiontypes	4876	1.89
addressline1	1817	0.71
addresstitle	5	0.0

4.3 openchargemap_charging_stations_sample

- Rows: 133026
- Columns: 17
- Duplicate rows: 223

Column names:

id, uuid, station_title, operator, usage_type, status, country, iso_code, state_or_province, town, address_line_1, postcode, latitude, longitude, number_of_connections, date_created, date_last_verified

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
date_last_verified	133026	100.0
state_or_province	51494	38.71
usage_type	29554	22.22
operator	23750	17.85
status	21610	16.24
postcode	10907	8.2
town	8500	6.39
address_line_1	70	0.05
id	0	0.0
iso_code	0	0.0

4.4 tesla_deliveries_2015_2025_raw

- Rows: 2640
- Columns: 12
- Duplicate rows: 0

Column names:

year, month, region, model, estimated_deliveries, production_units, avg_price_usd, battery_capacity_kwh, range_km, co2_saved_tons, source_type, charging_stations

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
year	0	0.0
month	0	0.0

Field	Missing Count	Missing Rate (%)
region	0	0.0
model	0	0.0
estimated_deliveries	0	0.0
production_units	0	0.0
avg_price_usd	0	0.0
battery_capacity_kwh	0	0.0
range_km	0	0.0
co2_saved_tons	0	0.0

4.5 us_alt_fuel_stations_historical_raw

- Rows: 95300
- Columns: 75
- Duplicate rows: 0

Column names:

fuel_type_code, station_name, street_address, intersection_directions, city, state, zip, plus4, station_phone, status_code, expected_date, groups_with_access_code, access_days_time, cards_accepted, bd_blends, ng_fill_type_code, ng_psi, ev_level1_evse_num, ev_level2_evse_num, ev_dc_fast_count, ev_other_info, ev_network, ev_network_web, geocode_status, latitude, longitude, date_last_confirmed, id, updated_at, owner_type_code, federal_agency_id, federal_agency_name, open_date, hydrogen_status_link, ng_vehicle_class, lpg_primary, e85_blender_pump, ev_connector_types, country, intersection_directions_french, access_days_time_french, bd_blends_french, groups_with_access_code_french, hydrogen_is_retail, access_code, access_detail_code, federal_agency_code, facility_type, cng_dispenser_num, cng_on_site_renewable_source, cng_total_compression_capacity, cng_storage_capacity, lng_on_site_renewable_source, e85_other_ethanol_blends, ev_pricing, ev_pricing_french, lpg_nozzle_types, hydrogen_pressures, hydrogen_standards, cng_fill_type_code, cng_psi, cng_vehicle_class, lng_vehicle_class, ev_on_site_renewable_source, restricted_access, rd_blends, rd_blends_french, rd_blended_with_biodiesel, rd_maximum_biodiesel_level, nps_unit_name, cng_station_sells_renewable_natural_gas, lng_station_sells_renewable_natural_gas, maximum_vehicle_class, ev_workplace_charging, funding_sources

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
expected_date	95300	100.0
plus4	95300	100.0
cng_storage_capacity	95300	100.0

Field	Missing Count	Missing Rate (%)
hydrogen_status_link	95300	100.0
ng_vehicle_class	95300	100.0
e85_blender_pump	95300	100.0
lpg_primary	95300	100.0
bd_blends_french	95300	100.0
ng_psi	95300	100.0
ng_fill_type_code	95300	100.0

4.6 world_bank_gdp_country_raw

- Rows: 266
- Columns: 71
- Duplicate rows: 0

Column names:

country_name, country_code, indicator_name, indicator_code, 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968, 1969, 1970, 1971, 1972, 1973, 1974, 1975, 1976, 1977, 1978, 1979, 1980, 1981, 1982, 1983, 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, unnamed:_70

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
unnamed:_70	266	100.0
2025	266	100.0
1960	115	43.23
1961	112	42.11
1963	110	41.35
1964	110	41.35
1962	110	41.35
1965	104	39.1
1966	103	38.72
1967	99	37.22

4.7 world_bank_income_level_country_raw

- Rows: 265
- Columns: 6
- Duplicate rows: 0

Column names:

country_code, region, incomegroup, specialnotes, tablename, unnamed:_5

Key field check:

- All predefined key fields are present.

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
unnamed:_5	265	100.0
specialnotes	132	49.81
incomegroup	50	18.87
region	48	18.11
country_code	0	0.0
tablename	0	0.0

4.8 world_bank_population_raw

- Rows: 17191
- Columns: 41
- Duplicate rows: 0

Column names:

datastructure, wb_data360:ds_data3601_3, i, a, vnm, wb_wdi_sp_pop_totl, _t, _t_1, _t_2, ps, _z, _z_1, _z_2, 1999, 7_6287452e7, _z_3, count, 2, wb_wdi, p1y, 0, a_1, pu, annual, vietnam, population,_total, total, all_age_ranges_or_no_breakdown_by_age, total_1, persons, not_applicable, not_applicable_1, not_applicable_2, not_applicable_3, count_integer, two, world_development_indicators_wdi, annual_1, units, normal_value, public

Key field check:

- Missing key fields: country_name, country_code, indicator_name, indicator_code

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
datastructure	0	0.0
wb_data360:ds_data3601_3	0	0.0

Field	Missing Count	Missing Rate (%)
i	0	0.0
a	0	0.0
vnm	0	0.0
wb_wdi_sp_pop_totl	0	0.0
_t	0	0.0
_t_1	0	0.0
_t_2	0	0.0
ps	0	0.0

4.9 world_bank_road_density_raw

- Rows: 1103
- Columns: 38
- Duplicate rows: 0

Column names:

datastructure, wb_data360:ds_data3601_2, i, a, annual, alb, albania, wef_ttdi_roaddens, road_density, km_surface_area, _t, total, _t_1, all_age_ranges_or_no_breakdown_by_age, _t_2, total_1, km, kilometers, wef_ttdi_val, type:_value, _z, not_applicable, z_1, *not_applicable_1*, 2021, 68_03394160583942, *wef_ttdi*, *travel&_tourism_development_index_ttdi*, 0, units, number, number_real_number, 602, ccyy, euromonitor_road_network_data_divided_by_land_area, a_1, normal_value, pu, public

Top missing-value fields:

Field	Missing Count	Missing Rate (%)
euromonitor_road_network_data_divided_by_land_area	36	3.26
68_03394160583942	12	1.09
i	0	0.0
a	0	0.0
datastructure	0	0.0
wb_data360:ds_data3601_2	0	0.0
alb	0	0.0
annual	0	0.0
albania	0	0.0
wef_ttdi_roaddens	0	0.0

5. Initial Findings

- Multiple raw data sources have been successfully collected and loaded into the SQLite database.
- Charging station datasets contain useful geographic fields such as latitude and longitude, which will support spatial analysis in later stages.
- Some datasets contain missing values in operator, usage type, status, pricing, or metadata fields. These fields should be reviewed during Week 2 data cleaning.
- Some World Bank datasets have different structures depending on the indicator source. These files may require reshaping into a consistent country-year format.
- A small number of duplicate rows exist in selected raw tables. Duplicate station checks should use station name, operator, address, latitude, and longitude instead of exact-row matching only.
- Dataset merging should not be performed before standardization. Standardization, deduplication, and integration should be conducted in Week 2.

6. Next Steps

- Standardize country names and country codes across charging station, EV sales, and external factor datasets.
- Convert World Bank data into a more consistent country-year format when needed.
- Identify Tesla-related charging stations from broader charging station datasets.
- Check duplicate charging stations using station name, operator, address, latitude, and longitude.
- Build a cleaned dataset for exploratory data analysis in Week 2.